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* Unit I: Overview of Operating System [8M]

• Operating System

(i) Definition: An operating system is a program which acts as an interface between computer hardware and software.

(ii) eg: Windows, Linux, Android, Mac, Ubuntu, etc.

(iii) Goals/Needs:

- Execute user programs and make solving user problems easier.
- Makes the computer system convenient to use.
- Uses computer hardware in an efficient manner.

• Components of OS: (W-19) (4M) (S-22)

— There are 8 components of an operating system.

(i) Process Management:

- A program in running state is called a process. Process needs various system
- The process management component manages the multiple processes running simultaneously on OS.
- Activities of process management:

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- Mechanism for process synchronization & deadlock handling
 - Create, load, execute, suspend and terminate processes.
 - Switch system among multiple processes in main memory.
 - Provides communication mechanisms so that processes can communicate with each other.
 - Allocate/de-allocate resources properly to prevent deadlock situation.

ii) I/O device management:

a) They provide an abstract level of hardware devices and keep details from applications to ensure proper use of devices, to prevent errors & to provide users with convenient & efficient environment. Some examples of storage media are magnetic tape, magnetic disk.

b) Tasks:

- Hide details of hardware devices
- Manage main memory for devices using cache, buffer and spooling
- Maintain & provide custom drivers for each device.

iii) File management:

a) It is one of the most visible services of an operating system.

b) Computers can store information in several different physical forms such as magnetic tape, disk, and drum are most common forms.

c) Tasks:

- File creation and deletion
- Directory creation and deletion
- File backup on stable (nonvolatile) storage media.
- Mapping files onto secondary storage

iv) Main memory management:

a) Memory is a large array of words or bytes, each with its own address. Main memory is a volatile storage device which means it loses its contents in case of system failure or as soon as system power goes down.

b) The main motivation behind memory management is to maximize memory utilization on computer system.

c) Tasks:

- Keep track of which parts of memory are currently being used and by whom.
- Decide which processes to load when memory space becomes available.
- Allocate & deallocate memory space as needed.

v) Secondary storage management:

a) Main purpose of a computer system is to execute programs. Since the main memory is too small to permanently accommodate all data & program, computer system must provide secondary storage to backup main memory.

b) Most programs like compilers, assemblers, editors, formatters, etc. are stored on the disk until loaded into memory and then use the disk as both source & destination of their processing.

c) Tasks:

- Free space management
- Storage allocation
- Disk scheduling

vi) Network Management:

a) It is the process of keeping your network healthy for an efficient communication between different computers.

b) It comprises fault analysis, maintaining quality of service, performance management.

c) Features:

- Network maintenance
- Network administration
- Network security

▪ Operations of OS / Functions / Services

① Memory management

② Device management

③ Processor management

④ Security

⑤ Error detection

⑥ Coordination between S/W & user

⑦ Job accounting

⑧ File management.

A] Memory management:

- Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
- Allocates the memory when a process requests & De-allocates the memory when a process no longer needs it.

B] Processor management:

- Keeps tracks of processor and status of process. The program responsible for this task is known as traffic controller.
- Allocates processor to a process & De-allocates processor when process is no longer required.

C] Device management:

- Keeps tracks of all devices. The program responsible for this task is known as I/O controller.
- Decides which process gets the device when and for how much time.
- Allocates & De-allocates device in most efficient way.

D] File management:

- Keeps track of information, location, uses, status, etc. and decides who gets the resources.
- Allocates & De-allocates resources.

E] Security:
 By means of password & similar other techniques, it prevents unauthorized access to programs & data.

F] Job accounting:
 Keeping track of time & resources used by various jobs & users.

G] Error detecting aids: (user)
 Production of dumps, traces, error messages and other debugging & error detecting aids.

H] Coordination between other s/w & users:
 Coordination & assignment of compilers, interpreters, assemblers & other software to various users of computer systems.

• Views of OS

i) The operating system can be observed from the point of view of the user or system. This is known as user view and system view respectively.

ii) Types of views:

A] User view

— The user viewpoint focuses on how the user interacts with operating system through the usage of various application programs.

- An operating system is designed primarily for ease of use with little emphasis on quality and none of resource utilization.

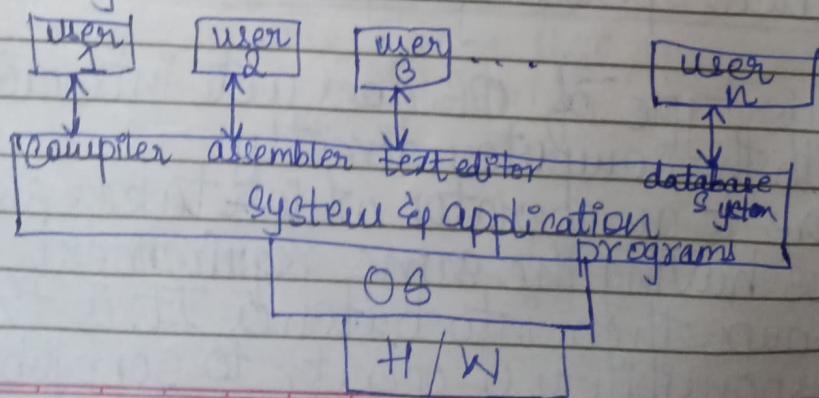
8] System View

- An operating system can also be considered as program running at all times in the background of a computer system known as kernel & handling all application programs. The operating system may also be viewed as just a resource allocator.
- In the system viewpoint the operating system is more involved with hardware services - CPU time, memory space, I/O operation, etc.

(iii) The user viewpoint is all about how user has to interact with operating system with help of various application programs.

(iv) From system point of view we see how hardware has to interact with operating system for accomplishing various tasks.

Diagram:



▪ Difference: Command Line Interface (CLI) & Graphical User Interface (GUI)

CLI

GUI

- | | |
|---|--|
| i) Allows user to perform tasks by issuing in lines of text or command lines. | It enables users to interact with the operating system or application. |
| ii) Less memory is required | More memory as it has various graphics components. |
| iii) Not Not easy to use | Easy to use |
| iv) Faster than GUI | Slower than CLI |
| v) Needs only keyboard | Both keyboard & mouse are needed. |
| vi) CLI CLI does not change, remains same over time | Structure & design can change with updates. |
| vii) eg: DOS, UNIX | eg: Windows, Linux |

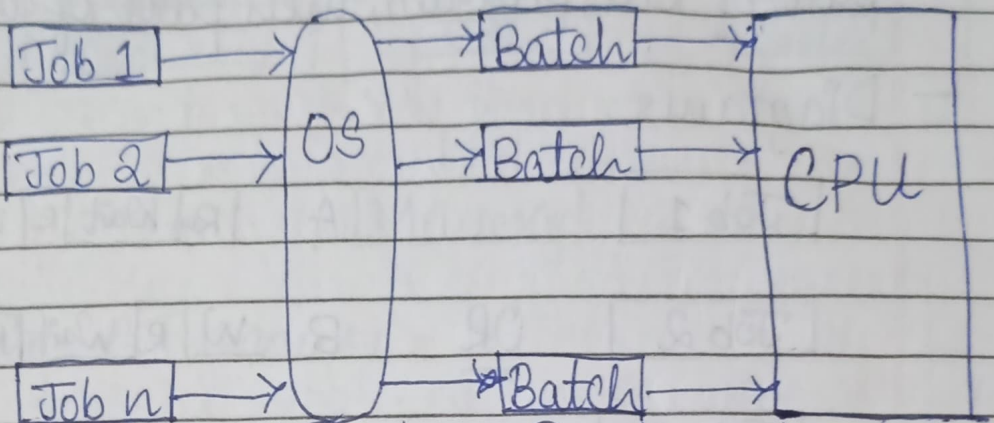
▪ Types of OS

i) Batch Operating System:

- This type of OS does not interact with the computer directly.
- There is an operator which takes similar jobs having the same requirement and groups them into batches. It is the responsibility of operator to sort jobs

with similar needs.

— Diagram:



— Eg: Payroll system, Bank system, etc.

— Advantages:

- i Multiple users can share the batch systems.
- ii Idle time is very less.
- iii Easy to manage large work repeatedly.

— Disadvantages:

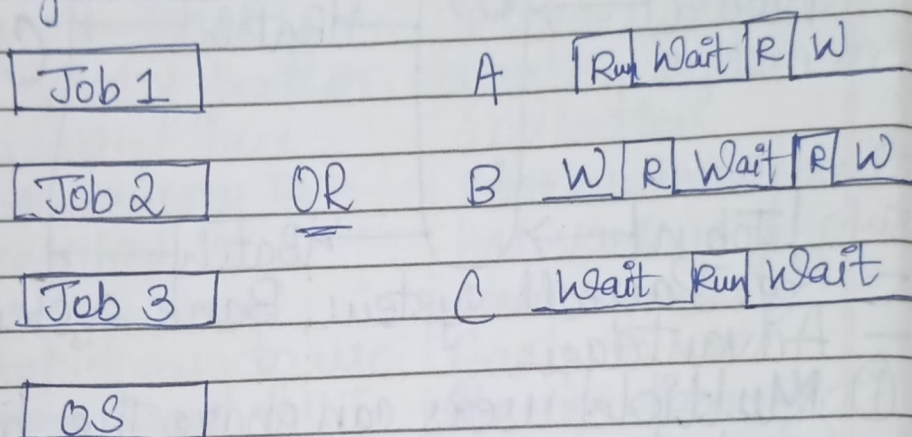
- i Debugging is difficult.
- ii Lack of interaction between user & job.
- iii Costly at times.

② Multi-Programmed OS:

- In this, more than one program lies in the memory. Scheduler selects the jobs to be placed in ready queue from number of programmes.
- Existence of more than one program in main memory is known as multi-programming.

- As there is only one processor, multiple programs cannot be executed at a time. So, operating system executes part of one program, then part of another.

- Diagram:

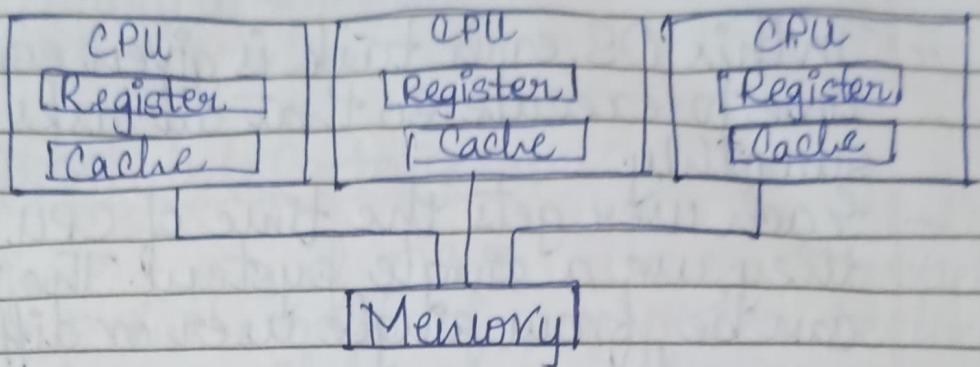


- Eg: User can open word, excel, access & other applications in a system.

③ Multi-Processing OS: (Parallel)

- In Multiprocessing, parallel computing is achieved. There are more than one processors present in system which can execute more than one process at same time.
- Increases the throughput of the system. & also reliability increases.
- The only disadvantage is it is more complex as it takes care of multiple CPUs simultaneously.

Diagram:



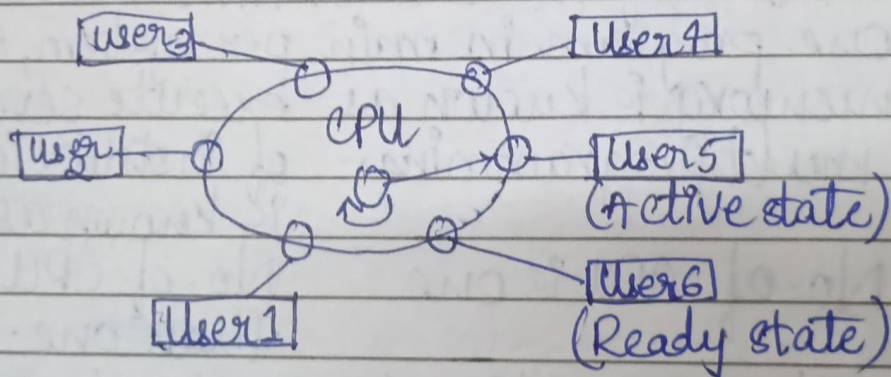
- e.g.: CT Scans.

★ Multi-Programmed Multi-Processing

- | | |
|---|--|
| <p>(i) The concurrent application of more than one program in main memory is known as multiprogramming.</p> | <p>The availability of more than one processor per system, that can execute several set of instructions, parallel is known as multiprocessing.</p> |
| <p>(ii) No. of CPU is one</p> | <p>No. of CPU is more than one.</p> |
| <p>(iii) Takes more time to process the jobs</p> | <p>Takes less time for job processing</p> |
| <p>(iv) More One process can be executed at a time.</p> | <p>More than one process can be executed at a time.</p> |
| <p>(v) No. of users is one at a time</p> | <p>No. of users can be one or more than one.</p> |
| <p>(vi) Throughput & Efficiency is less</p> | <p>Throughput & efficiency is maximum.</p> |

② Time Shared OS (3-22) (W-22) (4M)

- In this OS, each task is given some time to execute so that all tasks work smoothly.
- Each user gets the time of CPU as they use a single system. The task can be from single user or different users also.
- The time that each task gets to execute is called quantum. After this time interval is over, OS switches over next task.
- Diagram:



Here, User 1, 2, 3 & 4 are in waiting state as they are waiting for I/O operation. User 5's program is under execution & User 6's program is ready to execute & waiting for its next turn for CPU.

- Advantages:

- i) Each task gets an equal opportunity.
- ii) Fewer chances of duplication of s/w.

Eg: UNIX & Linux

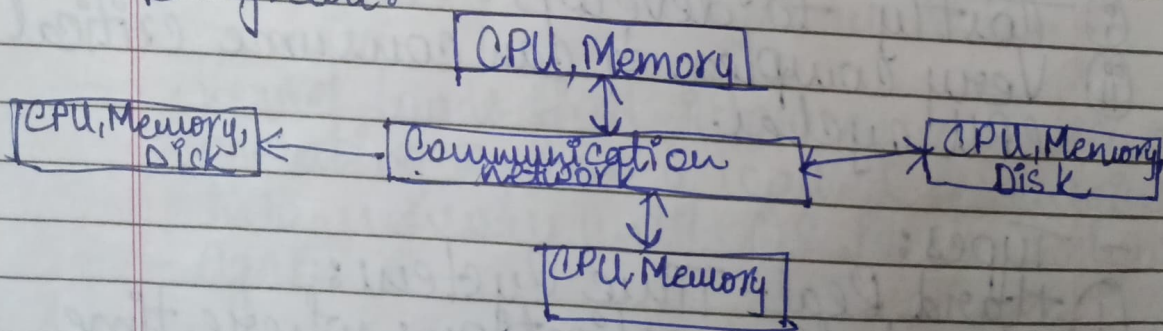
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- Disadvantages:
 - (i) Reliability problem
 - (ii) Security risks

⑤ Distributed Systems

- These types of OS is a recent advancement in world of computer technology and are widely accepted.
- Various autonomous interconnected computers communicate with each other using shared communication.
- Independent systems possess their own memory unit and CPU. They are referred as loosely coupled systems or distributed systems.
- Major benefit of working with these types of operating system is that it is always possible that one user can access the files which are not actually present on his system but some other system connected within network.
- Diagram:



⑥ Real-Time OS

- These type of OSs serve real-time systems. The time interval required to process & respond to inputs is very small. This interval is known as response time.
- They are used when there are time requirements that are very strict like robots, air traffic control systems, etc.
- In these systems, each job carries a certain deadline within which job is supposed to be completed, otherwise huge loss will be there or even if result is produced it will be useless.

- Advantages:

- i Easy to layout, develop & execute real-time applications under real-time OS.
- ii Maximum utilization of devices & systems.

- Disadvantages:

- i Costly to develop
- ii Very complex & can consume critical CPU cycles.

- Types:

- i Hard Real-Time Systems:

Meant for applications where time constraints are very strict & even shortest possible delay is not acceptable. Used in real-time systems.

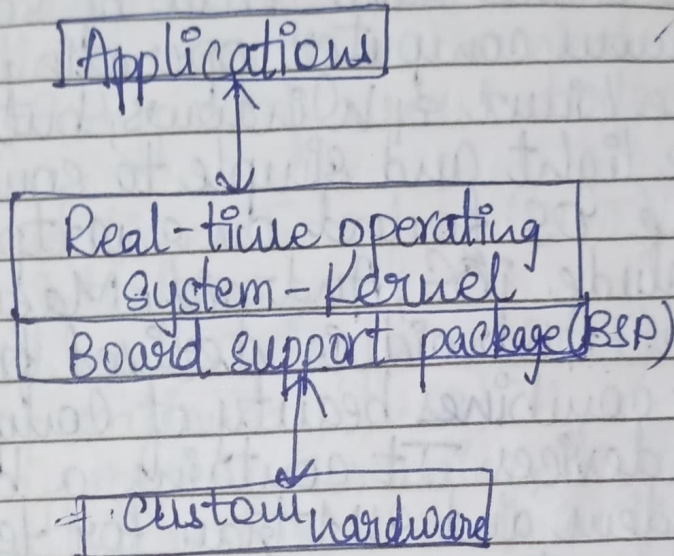
Applications of real-time OS

- i) Flight control system
- ii) Simulations
- iii) Industrial control
- iv) Military applications

② Soft Real-Time Systems:

These OSs are for applications where time-constraint is less strict. Used in digital audio systems.

- Diagram:



Kernel: It is a central component of OS that manages operations of computer & hardware. Core component of an OS. Acts as a bridge between applications & data processing performed at hardware level using inter-process communication & system calls.

Kernel loads first into memory when an operating system is loaded & remains into memory until OS is shut down again.

⑦ Mobile OS

- A mobile OS is an operating system that helps to run other application software on mobile devices.
- It is the same kind of software as famous computer operating systems like Linux & Windows, but now they are light and simple to some extent.
- The OS found on smartphones include iOS, Android, Mac OS, WebOS etc. Android is derived from Linux.
- It combines beauty of computer & hand use devices. It contains a built-in modem and SIM tray for telephony & internet connections.

iOS

- i) Owned by Apple Development Incorporation
- ii) Launched in 2007
- iii) Specially designed for Apple iPhones & iPads.
- iv) Swift is majorly used for iOS application development.

Android

- Owned by Google LLC.
- Launched in 2008
- Designed for smartphones of all companies.
- Java & Kotlin are majorly used for Android application development.

Multi-Programming

- (i) When multiple programs execute at a time on single device, it is multiprogramming
- (ii) Concept of context switching is used
- (iii) Execution of process takes more time
- (iv) Useful for reducing/decreasing CPU idle time and increasing throughput as much as possible

Multi-Tasking

When a single resource is used to process multiple tasks then it is multitasking. Concept of time sharing is used. Execution of process takes less time. Useful for running multiple processes at same time, effectively increasing CPU & system throughput.